

WZIP and WLZ4: LZ77 Codecs with Match-Length-Dependent Sliding Windows

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Abstract

An LZ77 codec bounds every match by one sliding window, trading cheap offsets for short matches against reach for long ones. Match-length-dependent windows give finer control over this trade-off: each match length has its own window, and the decoder infers the offset range from the decoded length. We present two such codecs: WLZ4, a byte-aligned LZ4-class format with one-byte and flagged two- or three-byte offsets, and WZIP, an entropy-coded Zstandard-class format. Each compressed block begins with its decoded length, from which WZIP derives its widest window and a decoder sizes its output. LZ4 blocks may expand incompressible input by 0.4%, whereas WLZ4 adds at most 15 bytes and WZIP 2. We describe the formats, match finders, decoders and optimal parser. On Silesia, WLZ4 reaches a compression ratio up to 16% above LZ4HC's at a lower decoding speed, and WZIP exceeds Zstandard 22's ratio while decoding at 84–86% of its speed.

1 Introduction

LZ77 [1] replaces a repeated substring by a pair (ℓ, d) : copy ℓ bytes from d bytes back. Every practical LZ77 codec bounds d by a window W , and the choice of W is a compromise: a wide window exposes distant repetitions but makes every offset expensive, so that a 3-byte match within a 64 KB window saves little or nothing, while a narrow window misses long repetitions further back. Byte-aligned codecs such as LZ4 [2] fix the window at 64 KB and the minimum match at 4 bytes; entropy-coded codecs such as DEFLATE [3], Zstandard [4], Brotli [5] and LZMA/xz [6, 7] ease the compromise with variable-length offset codes, and LZMA and Brotli condition them on the match length (2, 3, 4, 5+), which changes the offsets' probabilities but not their range: every length still reaches the whole window.

The top of each class is hard to move. Zstandard at level 22 combines a 128 MB window, a binary-tree optimal parser and FSE entropy coding [8], refined over a decade (its `--max` mode, new in version 1.5.7, sets every parameter to its maximum and compresses enwik9 5.6% more at 12 times the time and 6 times the memory [9]); the codecs we measured that compress Silesia further, Brotli at level 11 and xz at $-9e$, rely on context-modeled literal coding and decode 2.5 and 8 times slower. LZ4HC at level 12 is similarly the strongest byte-aligned parser at LZ4's decoding speed. Our aim is to improve on both while keeping each class's decoding structure: WZIP, with Huffman coding only, exceeds the ratio of Zstandard 22 by 0.9–1.1% on Silesia at levels 11–12 while decoding at 84–86% of its speed (by 1.3% at level 13, decoding 22%

slower), and by 1.5–1.8% on the 1 GB enwik9, where it also exceeds xz but decodes 27–28% slower than Zstandard; WLZ4 reaches ratios up to 16% above LZ4HC 12’s on Silesia and 28% on enwik9, decoding 14–25% slower than LZ4HC 12 on Silesia and up to 45% on enwik9 (Section 6).

Match-length-dependent windows assign each length ℓ a window $W(\ell)$, nondecreasing in ℓ . Short matches are admitted only at short distances, where their offsets are cheap, and long matches reach the whole history; since lengths are decoded before offsets, the decoder knows $W(\ell)$. Byte-aligned formats have tied offset fields to lengths before: Snappy [10] limits its one-byte-offset copies (11-bit offsets) to lengths 4–11, and Lizard [11] gives 16-bit offsets to matches of 4 bytes or more and 24-bit offsets only to matches of 16 or more. WLZ4 instead gives lengths 3–4 a one-byte offset and lets a match of 4 bytes or more reach 8 MB for one extra offset byte, signalled by a flag rather than reserved to long matches. Our earlier abstract [12] studied a non-decreasing schedule over all lengths with an LZ4-based prototype, and [13] analyzes it: exact greedy parsing under length-dependent windows remains universal when the full window starts at a length $o(\log W)$, and a calibrated schedule never raises the minimum token cost over a single window. This paper is about engineering: two complete codecs, the design decisions the scheme enables, and measurements against current codecs. New relative to [12] are the flagged offset, WZIP’s size-derived schedule and joint symbols, the optimal parser, and the evaluation. Sections 2–4 describe the scheme and both formats, Section 5 their parser, and Section 6 compares them with LZ4 1.10, Zstandard 1.5.6, Brotli 1.1.0 and xz 5.6.2.

2 Length-dependent windows

A match (ℓ, d) at input position i copies $x_{i-d}, \dots, x_{i-d+\ell-1}$ to $x_i, \dots, x_{i+\ell-1}$, possibly overlapping itself. Under a schedule $W(\ell)$, (ℓ, d) is admissible if $d \leq W(\ell)$; a constant W is the single-window case. The schedule binds matches with a new offset: a repeat offset, coded by its cache slot, and WZIP’s run token are not restricted. In a byte-aligned format the schedule appears as offset fields of different sizes, selected in WLZ4 by the length code and a flag, in an entropy-coded format as offset alphabets that grow with the length; either way the decoder needs no extra memory. Both codecs start at length 3: a length-2 match costs WLZ4 a token and an offset byte, as much as the two literals it replaces, and added to WZIP_S (Section 4) within 16 bytes it lowered the ratio of 4–8 KB blocks by 0.01–0.08%.

The input length in the stream. A schedule may also depend on the input length n : a window wider than the input is wasted, and one that covers it lets the longest matches reach everything. Both formats therefore begin with n (two bytes below 32 KB, four otherwise; a stored WZIP input begins with a zero instead). The size costs at most four bytes and serves three purposes: WZIP derives a size-adapted base schedule $W_n(\ell)$ from it; the decoder allocates its output, plus a fixed slack, before decoding; and decoding can be checked against a known end. Snappy also stores the size up front [10]; the LZ4 block format omits it, LZ4 and Zstandard frames record it optionally, xz in its index, gzip modulo 2^{32} in a trailer, and Brotli not at all.

The schedule also shapes the search [12]: each class of lengths gets its own index,

Table 1: WLZ4 match codes (low nibble of the token) and offset fields; offset 0 ends a block.

Code	Length	Offset field	Window
0, 1	3, 4	1 byte	256 B
2–14	4–16	flagged: 2 bytes (flag 0) or 3 bytes (flag 1)	32 KB or 8 MB
15	17+ext.	flagged: 2 or 3 bytes	32 KB or 8 MB

keyed by a hash of the class’s minimum length and covering only the class’s window. Indexes on short seeds stay short even in repetitive data because their windows are short, long matches come from an index on long seeds over the whole window, and each index searched nearest first yields the nearest offset for each length, the cheapest when offset fields have fixed widths.

3 WLZ4: a byte-aligned multi-window format

Format. A block starts with its decoded size (2 bytes below 32 KB, else 4), so that, unlike an LZ4 block, it decodes without the size being supplied by the application. It holds sequences, each a token, a literal run and a match, as in LZ4. The token’s high nibble is the literal-run length; its low nibble is a match code (Table 1). Codes 0–1 are lengths 3–4 with a one-byte offset; codes 2–14 are lengths 4–16 and code 15 is length 17 plus an extension, with a *flagged offset*: a little-endian field whose low bit selects two bytes (a 15-bit offset, window 32 KB) or three (a 23-bit offset, window 8 MB). A nibble value of 15 announces an extension e : one byte if $e < 252$, else $251 + m$ followed by $e - 252$ in m bytes. A zero offset ends the block.

The flag is what makes the far window pay: an earlier variant spent match code 15 on far matches, which then always needed a length extension, and compressed Silesia 2.6–2.8% worse (Section 6). The extension code keeps the size of an extension logarithmic, and every encoder writes a sequence only if the output stays within four bytes of the input it covers, so incompressible input expands by at most 15 bytes, including the header, where an LZ4 block may add $n/255$, or 0.4%.

Decoding. The decoder reads four bytes v at the offset position; for codes 2–15 the offset size is $2 + (v \wedge 1)$ and the offset $v \gg 1$, masked, without a branch. Because the flag makes the position of the next token depend on a byte just read, the decoder takes the next token from v itself whenever no extension follows, which keeps a second load off each sequence’s critical path. Matches of up to 16 bytes at offsets of 8 or more take two 8-byte copies, longer ones 32-byte copies, and offsets below 8 pattern copies, as in LZ4’s fast loop.

Encoders. The fast and lazy levels probe two hash tables (3- and 5-byte hashes) and take a candidate of at least 3 bytes within 256 B, 5 bytes within 32 KB, or 6 bytes within 8 MB. The hash-chain levels 0–7 walk a 5-byte chain over 64 KB, compare candidates by net gain (length less the extra byte of a three-byte offset), and probe two far positions: the uncapped previous position of the same 5-byte hash, and the previous position of the same 8 bytes from a table sized to the input (at most 2^{20}

entries; it alone moved level 6 on Silesia from 2.70 to 2.81). Levels 8–12 run the optimal parser of Section 5 over a 3-byte chain (256 B), a 4-byte chain (64 KB) and a 6-byte chain (8 MB).

4 WZIP: an entropy-coded multi-window format

Format. WZIP codes an input of 32 KB or more as one block; smaller inputs (7 of the 29 small files) go to WZIP_M, a variant with fixed 8 and 32 KB windows and two interleaved sequence streams, and the repository adds WZIP_S for 4–8 KB storage pages. It starts with the decoded size n , which fixes the base schedule and its widest window, not a compression parameter as Brotli’s window bits or xz’s dictionary size are; a three-byte header then gives the widths the encoder chose for lengths 3–7, as gaps below the widest. The schedule has entries for lengths 3, 4, 5, 6, 7 and 8+: the power of two of n selects the base widths of the short lengths, and the widest window is raised to cover the whole input, up to 2^{27} bytes for lengths 8+ and 2^{26} for shorter ones. The longest matches thus reach every earlier byte of any input up to 128 MB at no cost in decoder memory, since a one-shot decoder holds the whole output anyway, while small inputs keep small windows and offset alphabets. For a 16 MB input the base widths are 12, 16, 20, 22, 25 and 25 bits; the optimal levels, whose parser prices each offset by its code length, widen lengths 3–5 to 18, 22 and 24 bits (and length 6 to 24, keeping the schedule nondecreasing). Literals are Huffman-coded [14] in their own stream; sequences, in blocks of 2^{14} , are coded as (i) a *joint symbol* of the offset-cache slot (four recent offsets or a new one), the literal-run class (0, 1, or 2+ literals) and the length symbol (lengths 3–4095 with extra bits, and a run token), 1020 values in all, or a 204-value alphabet without the slot, chosen per block; (ii) a literal-run symbol for runs of two or more; and (iii) for a new offset, a symbol from the Huffman table of the length’s group (lengths 3, 4, 5 and 6+, or eight finer groups at the highest level), plus raw bits. The cache is kept in move-to-front order. The run token codes a run of the preceding byte with its count in the offset field, bypassing the cache. The joint symbol captures the dependence of slot, class and length that separate Huffman codes, with their one-bit floor, miss. An input that WZIP cannot shrink by at least 32 bytes is stored after a zero size field, so no input grows by more than 2 bytes.

Encoders and decoder. Level 0 probes one hash table on 6-byte seeds in a 1 MB window; levels 1–3 are greedy and 4–6 lazy, comparing candidates by an estimated gain (four bits per byte less $\log_2 d$). Levels 7–13 run the optimal parser with three indexes matching the schedule: a 3-byte chain for lengths 3–4 searched to $W(4)$, a 5-byte chain for lengths 5–6 searched to $W(6)$, and a binary tree on 7-byte seeds over the whole window; each chain stops at the first match long enough for the next index, and after 16 steps.

5 Optimal parsing with length-dependent windows

Both codecs use the parser of Figure 1, an approximate shortest path over input segments like the “optimal” parsers of LZ4HC and Zstandard. Four aspects differ.

Parse a segment from position p , with carried literal run r_0 and offset cache R_0 . A node (c, s) is the cheapest known path from p to $p + c$ in state s ; it stores price, last step, previous state, pending literal run and (WZIP) offset cache. A pending run r is in state $\sigma(r) = 0$, or (WZIP levels 12–13) in its literal-run class $\sigma(r) = \min(r, 2)$. Initialize $(0, \sigma(r_0))$. For $c = 0, 1, \dots$:

1. if $c > 0$, for each state s at $c - 1$, with pending run r , relax $(c, \sigma(r + 1))$ by literal x_{p+c-1} and the change in the literal-run code;
2. stop at the last position reached (if $c > 0$), segment limit 4096 or input end;
3. gather candidates at $p + c$: per-class matches, and (WZIP) each state’s cached offsets and a run;
4. for each state s , candidate (L, d) and length $\ell \leq L$ that d may code, relax $(c + \ell, 0)$ with $\text{price}(c, s) + \text{cost}(\ell, d \mid s)$; (WZIP) its cache is the predecessor’s with d moved to the front, unchanged for a run;
5. on a candidate of the sufficient length, take it from the cheapest state and stop.

Back-trace from the cheapest state at the end, emit, and continue from there.

Figure 1: The optimal parser of WLZ4 (levels 8–12) and WZIP (levels 7–13).

Candidates. A single-window parser can price every length up to the longest match with the nearest candidate reaching it, the cheapest if offset prices do not decrease with distance. Under a schedule this fails twice. First, a candidate (L, d) codes only the lengths $\ell \leq L$ with $d \leq W(\ell)$: a far candidate may not code its own prefixes. WZIP sorts the matches of its three indexes by offset, keeps each only if it is longer than every nearer one and admissible at its own length, and prices each length with the nearest such candidate that admits it, skipping lengths no admissible candidate reaches. Second, in a byte format the offset cost steps with the distance, so WLZ4 keeps one candidate per offset class: the longest match within 256 B, within 32 KB and within 64 KB (one nearest-first chain walk that snapshots its best when crossing 32 KB), and beyond 64 KB, pricing each length with the cheapest. Cached offsets are coded by slot, not bound by the schedule, so WZIP also tries each cached offset at every position with every length, and a run, even of two bytes, as a run token.

Prices. WLZ4 prices are exact bytes; each node’s pending literal run is charged when the run ends. WZIP prices symbols in 1/256 bit from smoothed running frequencies, refreshed every 32 KB: a match costs its joint symbol, extra bits, and for a new offset its group’s offset symbol and raw bits; the literal-run code is charged incrementally. Because the parser’s output feeds its prices, the initial counts matter: a steep prior over lengths makes short lengths cheap enough that the parser splits long matches, and the resulting counts keep them cheap. WZIP therefore starts lengths from a weak prior that halves every eight symbols.

States. The joint symbol codes the literal-run class, so a match’s cost depends on how its path arrived: after a match, one literal, or more. Keeping one path per

position commits to one class and misprices the others. From level 12 WZIP keeps three nodes per position, one per class (a literal moves class s to $\min(s+1, 2)$, a match to 0), each with its own offset cache. This remains a bounded dynamic program, not a bit-optimal parse: paths that reach the same position and class with different caches or pending runs are merged, and segments, candidate limits and estimated prices restrict it further, as in the other “optimal” parsers of practical codecs.

Passes. Running prices lag behind changes in the data. At level 13 a first pass (shallow, one state) records the symbol counts of every 128 KB region, and the second pass prices each region from those counts: prices anticipate the local statistics that the second parse largely reproduces, one step of alternating between parse and code.

Relation to existing parsers. LZ4HC prices every length up to one longest match in exact bytes; Zstandard and LZMA [6] keep one path per position with its coder state, Zstandard seeding its strongest level with a trial pass over the first block; bit-optimal parsing [15] is exact for fixed costs. New here are per-class candidates with per-length admissibility, a state per literal-run class, and per-region first-pass prices.

6 Experiments

Setup. Intel Core i7-8850H (6 cores, 9 MB L3), 32 GB RAM, Windows 11; one thread pinned to one core. WLZ4, WZIP, LZ4 1.10.0 and Zstandard 1.5.6 were built from source with GCC 14.2 -O2; Brotli 1.1.0 (window 2^{24} , its standard maximum) and xz 5.6.2 (no check) are MSYS2 release builds. Each file is compressed whole, by one library call. Ratio is total input over total output; speeds are total input over total time after a one-second warm-up, best of 3 compressions for fast codecs (1 otherwise, the fast codecs being measured in two passes) and best of 10 decompressions; every file is verified. LZ4 decodes with its bounds-checked `LZ4_decompress_safe`; WLZ4 and WZIP decode in their opt-in trusted mode, needing no size from the caller as each stream stores it (their default, checked mode decodes Silesia 6–10% and at most 8% slower). Encoder memory is the growth of the peak working set during compression, not counting contexts created beforehand. Corpora: Silesia [16] (12 files, 5–51 MB, 211.9 MB); as small files, Canterbury and Calgary [17, 18] (29 files, 3.7 KB–1 MB, 6.0 MB); and as large inputs enwik8 and enwik9 [19], the first 10^8 and 10^9 bytes of an English Wikipedia dump, measured together in a later session.

The LZ4 class (Table 2, Figure 2). On Silesia every WLZ4 level from 2 up reaches a higher ratio than LZ4HC at its highest level: level 2 reaches 2.748 against 2.743 at 5.0 times LZ4HC-12’s compression speed, and the optimal levels 8, 10 and 12 reach 2.977, 3.095 and 3.179, up to 15.9% above it. The fast and lazy levels gain 12–15% over LZ4’s default mode at 40–46% of its compression speed. On the small files the far window has little to find, and the 32 KB two-byte window leaves the hash-chain levels about 2.3% below LZ4HC at similar speed, while levels 10 and 12 still exceed LZ4HC-12 (2.815 and 2.846, against 2.771). The price is decompression speed: WLZ4 decodes at 2.2–2.8 GB/s, against 2.9–3.2 GB/s for LZ4. Its encoder memory stays at most 40 MB, the optimal levels’ far chain over the 8 MB window.

The Zstandard class (Table 3). WZIP’s levels 0–9 lie close to Zstandard’s trade-off curve on Silesia but slightly below it (level 9: 3.944 at 2.5 MB/s; Zstandard 19:

Table 2: The LZ4 class: ratio, and compression and decompression speed (MB/s), on Silesia and on Canterbury+Calgary; encoder memory (MB) on Silesia.

Codec, level	Silesia			Canterbury+Calgary			Memory (MB)
	Ratio	Comp.	Dec.	Ratio	Comp.	Dec.	
LZ4 (default)	2.101	594	2882	2.074	544	3212	0.0
LZ4HC 9	2.721	29.5	3138	2.751	24.6	3228	0.3
LZ4HC 12	2.743	8.95	2991	2.771	4.19	3170	0.3
WLZ4 fast	2.360	272	2436	2.323	256	2339	0.0
WLZ4 lazy	2.423	240	2533	2.397	217	2483	0.0
WLZ4 2	2.748	44.9	2586	2.629	55.6	2667	4.4
WLZ4 6	2.820	27.6	2562	2.689	28.8	2805	4.4
WLZ4 8	2.977	6.84	2353	2.770	12.7	2588	40
WLZ4 10	3.095	4.15	2401	2.815	8.39	2612	40
WLZ4 12	3.179	1.16	2235	2.846	1.52	2624	40

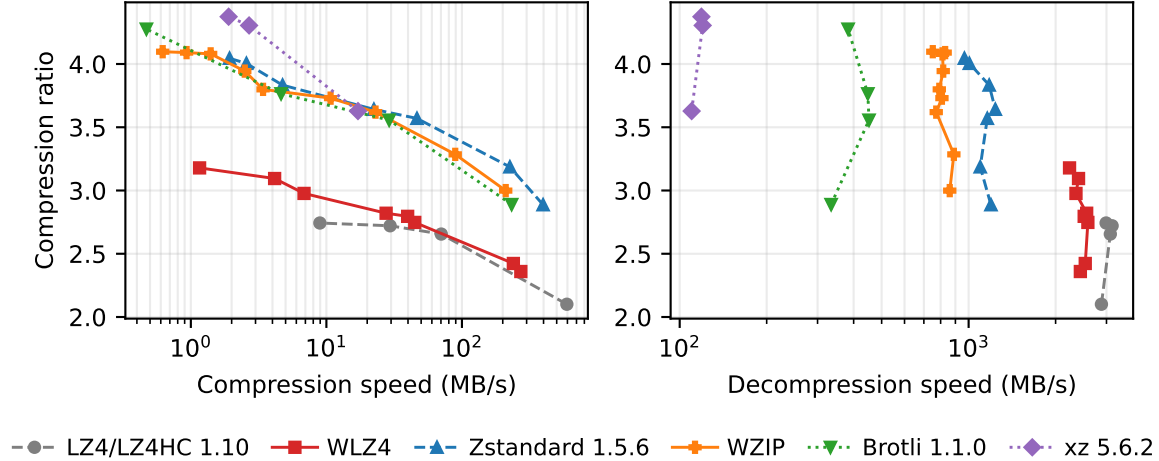


Figure 2: Silesia: compression ratio against compression speed (left) and decompression speed (right), log scale.

4.005 at 2.6 MB/s). Its top levels exceed Zstandard 22’s ratio: level 11 reaches 4.080, 0.9% above Zstandard 22’s 4.045, at 73% of its compression speed, and level 13 reaches 4.097 (+1.3%); on the small files level 13 exceeds Zstandard 22 by 0.8%. Zstandard 1.5.7 changes these ratios by at most 0.06% (level 22: 4.047 on Silesia, 4.210 on the small files, 3.957 on enwik8). WZIP decodes 14–28% slower than the Zstandard level of similar ratio in Table 3 (least at the top: levels 11 and 12 decode at 812 and 830 MB/s against Zstandard 22’s 967). Part of the gap may come from WZIP’s Huffman tables, several per block, where Zstandard decodes its sequences with FSE [4, 8]; the parsers and the decoders themselves also differ. WZIP decodes about twice as fast as Brotli 11 and 6.8 times as fast as xz, which compress Silesia further (4.276 and 4.374) at compression speeds below 2 MB/s. The encoder memory of WZIP’s optimal levels is about 950 MB on Silesia, mostly for the binary tree over a

Table 3: The Zstandard class: ratio, and compression and decompression speed (MB/s), on Silesia and on Canterbury+Calgary; encoder memory (MB) on Silesia.

Codec, level	Silesia			Canterbury+Calgary			Memory (MB)
	Ratio	Comp.	Dec.	Ratio	Comp.	Dec.	
Zstandard 1	2.887	397	1195	3.276	347	1181	0.5
Zstandard 3	3.186	226	1097	3.572	200	1026	1.3
Zstandard 19	4.005	2.57	1006	4.209	3.26	887	81
Zstandard 22	4.045	1.93	967	4.210	2.44	888	641
WZIP 0	2.999	210	860	3.378	166	827	9.6
WZIP 3	3.620	23.0	772	3.753	37.2	784	461
WZIP 9	3.944	2.50	817	4.147	4.12	760	947
WZIP 11	4.080	1.40	812	4.186	2.51	770	947
WZIP 12	4.090	0.93	830	4.195	1.40	784	947
WZIP 13	4.097	0.62	752	4.243	0.83	706	952
Brotli 11	4.276	0.47	383	4.490	0.56	390	213
xz -6	4.305	2.69	120	4.392	3.18	112	93
xz -9e	4.374	1.90	119	4.396	1.97	112	504

window that covers the whole input (up to 2^{27} positions); Zstandard 22’s is 641 MB.

Table 4: Large inputs: ratio, and compression and decompression speed (MB/s), on enwik8 (100 MB) and on enwik9 (1 GB); encoder memory (MB) on enwik9.

Codec, level	enwik8			enwik9			Memory (MB)
	Ratio	Comp.	Dec.	Ratio	Comp.	Dec.	
LZ4 (default)	1.746	486	2989	1.964	485	2729	0.0
LZ4HC 12	2.386	15.6	2561	2.691	16.6	2602	0.3
WLZ4 2	2.482	46.3	2094	2.812	47.2	2039	4.4
WLZ4 10	2.929	2.19	1458	3.332	2.62	1492	40
WLZ4 12	3.029	0.62	1514	3.451	0.75	1425	40
Zstandard 19	3.712	1.93	684	4.250	1.95	794	81
Zstandard 22	3.957	1.36	595	4.676	1.25	675	649
WZIP 9	3.793	1.37	562	4.408	1.47	545	2031
WZIP 11	3.970	0.97	488	4.745	0.92	492	2018
WZIP 13	3.983	0.45	470	4.759	0.42	484	2102
Brotli 11	3.885	0.45	358	4.477	0.47	386	204
xz -9e	4.027	1.32	108	4.722	1.34	136	673

Large inputs (Table 4). Our settings were tuned on Silesia and the small files, and WZIP’s windows for inputs above 64 MB on enwik8, the first tenth of enwik9, so enwik9 is not an independent test. WLZ4 12’s ratio is 27–28% above LZ4HC 12’s (level 10: 23–24%), but it decodes 41–45% slower, and level 2 exceeds LZ4HC 12 by 4% at about three times its compression speed. WZIP 11 exceeds Zstandard 22 by

0.3% on enwik8 and 1.5% on enwik9, and WZIP 13 by 0.7% and 1.8%; on enwik9, WZIP 13 compresses best of all the codecs tested, above xz $-9e$ (4.722) and Brotli 11 (4.477), and decodes 3.6 times as fast as xz but 28% slower than Zstandard 22. Its encoder memory is about 2 GB, three times Zstandard 22’s.

Ablation. Table 5 compares byte-aligned formats under optimal parsing (ours; LZ4HC-12’s for LZ4). On Silesia the far window adds 10–12% over two windows, and moving it from a match code to the flag adds 2.6–2.8% more, since far matches then need no length extension (the flag also halves both windows); on the small files, the flagged format’s 32 KB two-byte window costs 0.2–0.3%. Assigning offset sizes by length alone (e.g., 3 bytes from length 6), which needs neither code nor flag, lost 1–3% against the far code on Silesia and about 11% on the small files, where 47–64% of matches have length 6+ and lie within 64 KB.

WZIP’s parts, at level 11 on Silesia (4.080): coding the cache slot with the offset, not in the joint symbol (the 204-value alphabet throughout, same parse), lowers the ratio by 0.28%, and one offset table for all lengths instead of one per length group (reparsed) by 0.58%. Giving lengths 3–7 the widest window, with five offset tables in both runs, changes the ratio by +0.002% but on mozilla raises the peak working set from 1.06 to 1.44 GB and lowers the compression speed by 11%: in this configuration the gain lies in coding offsets per length, and the windows keep the indexes of short seeds small. Three parser states add 0.25% (level 12), and the second pass with eight offset groups 0.17% (level 13). The repository holds the switches and logs.

Table 5: Byte-aligned formats under optimal parsing: ratio at levels 10 and 12.

Offset fields	Silesia		Cant.+Calg.	
	L10	L12	L10	L12
LZ4: 2 B, 64 KB (LZ4HC-12’s parser)	—	2.743	—	2.771
1 B 256 B (len. 3–4); 2 B 64 KB	2.749	2.769	2.772	2.789
+ far code: 3 B 16 MB, length extension	3.015	3.091	2.822	2.854
WLZ4: 1 B 256 B; flagged 2 B 32 KB or 3 B 8 MB	3.095	3.179	2.815	2.846

7 Discussion and conclusion

Limitations. WLZ4 decodes 14–25% slower than LZ4HC 12 on Silesia, and up to 45% on enwik9, likely for three reasons: its parses contain more and shorter matches; the flag adds a data-dependent load to each sequence; and on large inputs a third of its matches copy from 64 KB to 8 MB back, beyond the L1 cache (37% of the output bytes at level 10 on Silesia). Since the encoder chooses the matches, a parse that also prices sequences and far copies for decoding time could trade some ratio for speed without changing the format.

Conclusion. Match-length-dependent windows let an LZ77 format give short matches cheap offsets without denying long matches a large history: as offset fields and a flag in a byte-aligned format, where the windows themselves raise the ratio, and as offset

alphabets coded per length group in an entropy-coded one, where under our optimal parser the gain comes from the per-length coding and the windows keep the search fast and small. Recording the input length up front costs at most four bytes and lets the schedule itself adapt to the input. Source code, the benchmark harness and scripts reproducing the tables are available at <https://github.com/icodywu/WZIP>.

References

- [1] J. Ziv and A. Lempel, “A universal algorithm for sequential data compression,” *IEEE Trans. Inf. Theory*, vol. 23, no. 3, pp. 337–343, 1977.
- [2] Y. Collet et al., “LZ4: extremely fast compression,” version 1.10.0, 2024. <https://github.com/lz4/lz4>.
- [3] P. Deutsch, “DEFLATE compressed data format specification version 1.3,” RFC 1951, 1996.
- [4] Y. Collet and M. Kucherauw, “Zstandard compression and the ‘application/zstd’ media type,” RFC 8878, 2021.
- [5] J. Alakuijala et al., “Brotli: A general-purpose data compressor,” *ACM Trans. Inf. Syst.*, vol. 37, no. 1, art. 4, 2019.
- [6] I. Pavlov, “LZMA specification (draft version),” 2015. <https://www.7-zip.org/sdk.html>.
- [7] L. Collin et al., “XZ Utils,” version 5.6.2, 2024. <https://tukaani.org/xz/>.
- [8] J. Duda, “Asymmetric numeral systems: entropy coding combining speed of Huffman coding with compression rate of arithmetic coding,” arXiv:1311.2540, 2013.
- [9] Zstandard v1.5.7 release notes, 2025. <https://github.com/facebook/zstd/releases/tag/v1.5.7>.
- [10] Google, “Snappy compressed format description,” 2011. https://github.com/google/snappy/blob/main/format_description.txt.
- [11] P. Skibiński, “Lizard v1.x block format,” 2016. <https://github.com/inikep/lizard>.
- [12] Y. Wu, “Improved LZ77 compression,” in *Proc. Data Compression Conf.*, 2021, p. 377.
- [13] Y. Wu, “Improved LZ77 compression with match-length-dependent sliding windows,” submitted to *IEEE Trans. Inf. Theory*, 2026. Preprint: <https://github.com/icodywu/WZIP/blob/main/papers/WLZ.pdf>.
- [14] D. A. Huffman, “A method for the construction of minimum-redundancy codes,” *Proc. IRE*, vol. 40, no. 9, pp. 1098–1101, 1952.
- [15] P. Ferragina, I. Nitto and R. Venturini, “On the bit-complexity of Lempel-Ziv compression,” *SIAM J. Comput.*, vol. 42, no. 4, pp. 1521–1541, 2013.
- [16] S. Deorowicz, “Universal lossless data compression algorithms,” Ph.D. dissertation, Silesian University of Technology, 2003.
- [17] R. Arnold and T. Bell, “A corpus for the evaluation of lossless compression algorithms,” in *Proc. Data Compression Conf.*, 1997, pp. 201–210.
- [18] T. C. Bell, J. G. Cleary and I. H. Witten, *Text Compression*. Prentice Hall, 1990.
- [19] M. Mahoney, “Large text compression benchmark,” 2006. <http://mattmahoney.net/dc/text.html>.